

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

RELATIONS & FUNCTIONS **PART-01 (01 to 70)**

DIBY (DO IT BY YOURSELF)

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DIBY-01

Consider set $A = \{-1, 0, 1, 2, 3\}$ and $B = \{1, 2, 4, 5, 0\}$, $R = \{(a, b) : b - a = 1, a \in A, b \in B\}$, then domain of relation R is:

- (A) $\{0, 1, 3\}$ (B) $\{-1, 0, 1, 3\}$ (C) $\{1, 0, -1, 2, 3\}$ (D) $\{1, 0, 2, 3\}$

[Ans. : (B)]

DIBY-02

Let $A = \{2, 4, 5, 7\}$ and $B = \{2, 4, 6\}$ be two sets and let a relation R be a relation from A to B is defined as $R = \{(x, y) : x < y, x \in A, y \in B\}$, then the difference between the sum of elements of domain and range of R is-

- (A) 2 (B) 3 (C) 4 (D) 1

[Ans. : (D)]

DIBY-03

If given $R = \{(x, y) : x, y \in W, x^2 + y^2 = 25\}$. Find the domain and range of R .

[Ans. : Domain of $R \equiv \{0, 3, 4, 5\}$
Range of $R \equiv \{5, 4, 3, 0\}$]

DIBY-04

Let $A = \{1, 2, 3, \dots, 10\}$ and $B = \left\{ \frac{m}{n} : m, n \in A, m < n \text{ and } \gcd(m, n) = 1 \right\}$. Then $n(B)$ is equal to
[JEE Mains-2025]

(A) 36

(B) 31

(C) 37

(D) 29

[Ans. : (B)]

DIBY-05

Let the number of elements in sets A and B be five and two respectively. Then the number of subsets of $A \times B$ each having at least 3 and at most 6 elements is:

[JEE Mains-2023]

(A) 752

(B) 782

(C) 792

(D) 772

[Ans. : (C)]

DIBY-06

The number of elements in the relation $R = \{(x, y) : 4x^2 + y^2 < 52, x, y \in \mathbb{Z}\}$ is:

[JEE Mains-2026]

- (A) 77 (B) 67 (C) 86 (D) 89

[Ans.: (A)]

DIBY-07

Let $A = \{2, 3, 6, 7\}$ and $B = \{4, 5, 6, 8\}$. Let R be a relation defined on $A \times B$ by $(a_1, b_1)R(a_2, b_2)$ is and if only if $a_1 + a_2 = b_1 + b_2$. Then the number of elements in R is

[JEE Mains-2024]

[Ans.: 25]

DIBY-08

Let $A = \{2, 3, 4\}$ and $B = \{8, 9, 12\}$. Then the number of elements in the relation

$R = \{((a_1, b_1)(a_2, b_2)) \in (A \times B, A \times B) : a_1 \text{ divides } b_2 \text{ and } a_2 \text{ divides } b_1\}$ is

[JEE Mains-2023]

- (A) 36 (B) 12 (C) 18 (D) 24

[Ans.: (A)]

DIBY-09

Let $A = \{1, 3, 4, 6, 9\}$ and $B = \{2, 4, 5, 8, 10\}$. Let R be a relation defined on $A \times B$ such that $R = \{((a_1, b_1), (a_2, b_2)) : a_1 \leq b_2 \text{ and } b_1 \leq a_2\}$. Then the number of elements in the set R is [JEE Mains-2023]

- (A) 26 (B) 160 (C) 180 (D) 52

[Ans.: (B)]

DIBY-10

Let $A = \{1, 2, 3, 4\}$ and R be a relation on the set $A \times A$ defined by $R = \{((a, b), (c, d)) : 2a + 3b = 4c + 5d\}$. Then the number of elements in R is

[JEE Mains-2023]

[Ans.: 6]

DIBY-11

Let $A = \{1, 2, 3, 4, \dots, 10\}$ and $B = \{0, 1, 2, 3, 4\}$. The number of elements in the relation $R = \{(a, b) \in A \times A : 2(a - b)^2 + 3(a - b) \in B\}$ is [JEE Mains-2023]

[Ans.: 18]

DIBY-12

If $A = \{a, b, c\}$, $B = \{b, c, d\}$ and $C = \{a, d, c\}$, then $(A - B) \times (B \cap C)$ is equal to

- (A) $\{(a, c), (a, d)\}$ (B) $\{(a, b), (c, d)\}$
 (C) $\{(c, a), (d, a)\}$ (D) $\{(a, c), (a, d), (b, d)\}$

[Ans.: (A)]

DIBY-13

If $A = \{3, 5, 7\}$ and $B = \{1, 2, 3, 5\}$, then $(A \times B) \cap (B \times A)$ is equal to

- (A) $\{(3, 3), (5, 3), (2, 7), (7, 2)\}$ (B) $\{(3, 3), (3, 5), (5, 3), (5, 5)\}$
 (C) $\{(3, 3), (5, 5)\}$ (D) $\{(3, 5), (5, 5), (5, 3)\}$

[Ans.: (B)]

DIBY-14

Define relation R_1 and R_2 on set $A = \{2, 3, 5, 7, 10\}$ as xR_1y if $x \mid (y - 1)$ and xR_2y of $(x + y = 10)$, then the relation R given by $R = R_1 \cap R_2$ is

- (A) $\{(2, 4)\}$ (B) $\{(3, 7)\}$ (C) $\{(3, 7), (5, 5)\}$ (D) None of these

[Ans.: (B)]

DIBY-15

Let the set $C = \{(x, y) \mid x^2 - 2^y = 2023, x, y \in \mathbb{N}\}$. Then $\sum_{(x,y) \in C} (x + y)$ is equal to

[JEE Mains-2024]

[Ans. : 46]

DIBY-16

Let $A = \{(x, y) \in R \times R : |x + y| \geq 3\}$ and $B = \{(x, y) \in R \times R : |x| + |y| \leq 3\}$

If $C = \{(x, y) \in A \cap B : x = 0 \text{ or } y = 0\}$, then $\sum_{(x,y) \in C} |x + y|$ is

[JEE Mains-2025]

(A) 18

(B) 12

(C) 24

(D) 15

[Ans. : (B)]

DIBY-17

Let $A = \{(\alpha, \beta) \in R \times R : |\alpha - 1| \leq 4 \text{ and } |\beta - 5| \leq 6\}$ and

$B = \{(\alpha, \beta) \in R \times R : 16(\alpha - 2)^2 + 9(\beta - 6)^2 \leq 144\}$. Then show that B is subset of A.



DIBY-18

Let $S = \{p_1, p_2, \dots, p_{10}\}$ be the set of first ten prime numbers. Let $A = S \cup P$, where P is the set of all possible products of distinct elements of S . Then the number of all ordered pairs $(x, y), x \in S, y \in A$, such that x divides y , is **[JEE Mains-2025]**
[Ans. : 5120]

DIBY-19

Let S be the set of the first 11 natural numbers. Then the number of elements in $A = \{B \subseteq S : n(B) \geq 2 \text{ and the product of all elements of } B \text{ is even}\}$ is _____. **[JEE Mains-2026]**
[Ans. : 1979]

DIBY-20

Let R_1 and R_2 be relations on the set $(1, 2, \dots, 50)$ such that $R_1 = \{(p, p^n) : p \text{ is a prime and } n \geq 0 \text{ is an intger}\}$ and $R_2 = \{(p, p^n) : p \text{ is a prime and } n = 0 \text{ or } 1\}$ Then, the number of elements in $R_1 - R_2$ is _____. **[JEE Mains-2022]**
[Ans. : 8]

TYPES OF RELATIONS

DIBY-21

Let R be a relation defined on $\mathbb{N}$ as aRb is $2a + 3b$ is a multiple of 5, $a, b \in \mathbb{N}$. Then R is [JEE Mains-2023]

- | | |
|----------------------------------|----------------------------------|
| (A) Not reflexive | (B) Transitive but not symmetric |
| (C) Symmetric but not transitive | (D) An equivalence relation |

[Ans. : (D)]

DIBY-22

Let $A = \{1, 2, 3, 4, 5, 6, 7\}$. Then the relation $R = \{(x, y) \in A \times A : x + y = 7\}$ is [JEE Mains-2023]

- (A) Transitive but neither symmetric nor reflexive
- (B) Reflexive but neither symmetric nor transitive
- (C) An equivalence relation
- (D) Symmetric but neither reflexive nor transitive

[Ans. : (D)]

TYPES OF RELATIONS

DIBY-23

Let R be a relation on $N \times N$ defined by $(a, b)R(c, d)$ if and only if $ad(b - c) = bc(a - d)$. Then R is

[JEE Mains-2023]

- (A) Symmetric but neither reflexive nor transitive.
- (B) Transitive but neither reflexive nor symmetric.
- (C) Reflexive and symmetric but not transitive.
- (D) Symmetric and transitive but not reflexive.

[Ans. : (A)]

DIBY-24

Let relation R on $\mathbb{N} \times \mathbb{N}$ be defined as : $(x_1, y_1)R(x_2, y_2)$ if and only if $x_1 \leq x_2$ or $y_1 \leq y_2$. Consider the two statements :

- (I) R is reflexive but not symmetric.
- (II) R is transitive

Then which one of the following is true?

[JEE Mains-2024]

- (A) Only (II) is correct.
- (B) Only (I) is correct.
- (C) Both (I) and (II) are correct.
- (D) Neither (I) nor (II) is correct.

[Ans. : (B)]

TYPES OF RELATIONS

DIBY-25

Consider the relations R_1 and R_2 defined as $aR_1b \Leftrightarrow a^2 + b^2 = 1$ for all $a, b \in R$ and $(a, b)R_2(c, d) \Leftrightarrow a + d = b + c$ for all $(a, b), (c, d) \in N \times N$. Then **[JEE Mains-2024]**

- (A) Only R_1 is an equivalence relation
- (B) Only R_2 is an equivalence relation
- (C) R_1 and R_2 both are equivalence relations
- (D) Neither R_1 nor R_2 is an equivalence relation

[Ans. : (B)]

DIBY-26

Let $A = \{-4, -3, -2, 0, 1, 3, 4\}$ and $R = \{(a, b) \in A \times A : b = |a| \text{ or } b^2 = a + 1\}$ be a relation on A . Then the minimum number of elements, that must be added to the relation R so that it becomes reflexive and symmetric, is **[JEE Mains-2023]**

[Ans. : 7]

TYPES OF RELATIONS

DIBY-27

The minimum number of elements that must be added to the relation

$R = \{(a, b), (b, c), (b, d)\}$ on the set $\{a, b, c, d\}$ so that it is an equivalence relation, is

[JEE Mains-2023]

[Ans. : 13]

DIBY-28

Let $A = \{1, 2, 3, 4\}$ and R be a relation on the set $A \times A$ defined by

$R = \{((a, b), (c, d)) : 2a + 3b = 4c + 5d\}$. Then the number of elements in R is:

[JEE Mains-2023]

[Ans. : 6]

TYPES OF RELATIONS

DIBY-29

Let R_1 and R_2 be two relations defined on R by $aR_1b \Leftrightarrow ab \geq 0$ and $aR_2b \Leftrightarrow a \geq b$

Then, [JEE Mains-2022]

- (A) R_1 is an equivalence relation but not R_2
- (B) R_2 is an equivalence relation but not R_1
- (C) Both R_1 and R_2 are equivalence relations
- (D) Neither R_1 nor R_2 is an equivalence relation

[Ans. : (D)]

DIBY-30

The minimum number of elements that must be added to the relation

$R = \{(a, b), (b, c)\}$ on the set $\{a, b, c\}$ so that it becomes symmetric and transitive is:

[JEE Mains-2023]

- (A) 4
- (B) 7
- (C) 5
- (D) 3

[Ans. : (B)]

TYPES OF RELATIONS

DIBY-31

Let $R_1 = \{(a, b) \in N \times N : |a - b| \leq 13\}$ and $R_2 = \{(a, b) \in N \times N : |a - b| \neq 13\}$. Then on N .
[JEE Mains-2022]

- (A) Both R_1 and R_2 are equivalence relations.
- (B) Neither R_1 nor R_2 is an equivalence relation.
- (C) R_1 is an equivalence relation but R_2 is not.
- (D) R_2 is an equivalence relation but R_1 is not.

[Ans. : (B)]

DIBY-32

Let R be a relation on $N \times N$ defined by $(a, b)R(c, d)$, if and only if $ad(b - c) = bc(a - d)$.
[JEE Mains-2023]

- (A) transitive but neither reflexive nor symmetric.
- (B) symmetric but neither reflexive nor transitive.
- (C) symmetric and transitive but not reflexive.
- (D) reflexive and symmetric but not transitive.

[Ans. : (B)]

TYPES OF RELATIONS

DIBY-33

Let $A = \{1, 2, 3, 4, 5\}$. Let R be a relation on A defined by xRy , if and only if $4x \leq 5y$. Let m be the number of elements in R and n be the minimum number of elements from $A \times A$ that are required to be added to R to make it a symmetric relation. Then $m + n$ is equal to

[JEE Mains-2024]

- (A) 25 (B) 24 (C) 26 (D) 23

[Ans.: (A)]

DIBY-34

Let W denotes the words in the English dictionary. Define the relation R by:

$R = \{(x, y) \in W \times W \mid \text{the words } x \text{ and } y \text{ have at least one letter in common}\}$. Then R is

[JEE Mains-2023]

- (A) Reflexive, symmetric and not transitive
 (B) Reflexive, symmetric and transitive
 (C) Reflexive, not symmetric and transitive
 (D) Not reflexive, symmetric and transitive

[Ans.: (A)]

TYPES OF RELATIONS

DIBY-35

If $R = \{(x, y) : x, y \in \mathbb{Z}, x^2 + 3y^2 \leq 8\}$ is a relation on the set of integers $\mathbb{Z}$, then the domain of R^{-1} is [JEE Mains-2020]

- (A) $\{0, 1\}$ (B) $\{-2, -1, 1, 2\}$ (C) $\{-1, 0, 1\}$ (D) $\{-2, -1, 0, 1, 2\}$
[Ans.: (C)]

DIBY-36

Which of the following is not correct for relation R on the set of real numbers? [JEE Mains-2021]

- (A) $(x, y) \in R \Leftrightarrow 0 < |x| - |y| \leq 1$ is neither transitive nor symmetric.
 (B) $(x, y) \in R \Leftrightarrow |x - y| \leq 1$ is reflexive and symmetric.
 (C) $(x, y) \in R \Leftrightarrow |x| - |y| \leq 1$ is reflexive but not symmetric.
 (D) $(x, y) \in R \Leftrightarrow 0 < |x - y| \leq 1$ is symmetric and transitive.

[Ans.: (D)]

TYPES OF RELATIONS

DIBY-37

Let a relation R on $N \times N$ be defined as : $(x_1, y_1)R(x_2, y_2)$ if and only if $x_1 \leq x_2$ or $y_1 \leq y_2$. Consider the two statements :

- (I) R is reflexive but not symmetric
- (II) R is transitive

Then which one of the following is true?

[JEE Mains-2024]

- (A) Both (I) and (II) are correct.
- (B) Only (II) is correct.
- (C) Neither (I) nor (II) is correct.
- (D) Only (I) is correct.

[Ans. : (D)]

DIBY-38

Let $A = \{0, 3, 4, 6, 7, 8, 9, 10\}$ and R be the relation defined on A such that

$R = \{(x, y) \in A \times A : x - y \text{ is odd positive integer or } x - y = 2\}$.

The minimum number of elements that must be added to the relation R , so that it is a symmetric relation, is equal to ____.

[JEE Mains-2023]

[Ans. : 19]

TYPES OF RELATIONS

DIBY-39

The relation $R = \{(x, y) : x, y \in \mathbb{Z} \text{ and } x + y \text{ is even}\}$ is:

- (A) reflexive and symmetric but not transitive.
- (B) an equivalence relation.
- (C) symmetric and transitive but not reflexive.
- (D) reflexive and transitive but not symmetric.

[JEE Mains-2025]

[Ans. : (B)]

DIBY-40

For $\alpha \in \mathbb{N}$, consider a relation R on $\mathbb{N}$ given by $R = \{(x, y) : 3x + \alpha y \text{ is a multiple of } 7\}$.

The relation R is an equivalence relation if and only if :

[JEE Mains-2022]

- (A) $\alpha = 14$
- (B) α is a multiple of 4
- (C) 4 is the remainder when α is divided by 10
- (D) 4 is the remainder when α is divided by 7

[Ans. : (D)]

TYPES OF RELATIONS

DIBY-41

Let a set $A = A_1 \cup A_2 \cup \dots \cup A_k$, where $A_i \cap A_j = \phi$ for $i \neq j, 1 \leq i, j \leq k$. Define the relation R from A to A by $R = \{(x, y): y \in A_i \text{ if and only if } x \in A_i, 1 \leq i \leq k\}$. Then, R is
[JEE Mains-2022]

- (A) Reflexive, symmetric but not transitive
- (B) Reflexive, transitive but not symmetric
- (C) Reflexive but not symmetric and transitive
- (D) An equivalence relation

[Ans.: (D)]

DIBY-42

The probability that a relation R from $\{x, y\}$ to $\{x, y\}$ is both symmetric and transitive, is equal to
[JEE Mains-2022]

- (A) $\frac{5}{16}$
- (B) $\frac{9}{16}$
- (C) $\frac{11}{16}$
- (D) $\frac{13}{16}$

[Ans.: (A)]



TYPES OF RELATIONS

DIBY-43

The number of symmetric relations defined on the set $\{1, 2, 3, 4\}$ which are not reflexive is_____.

[Ans. : 960]

DIBY-44

If $n(A) = 3$ and

l = number of Identity relations on set- A

m = number of reflexive relations on set- A

n = number of non-empty symmetric relations on set A

p = number of transitive relations on set A

Then value of $(l + m) - (n + p)$ is _____

[Ans. : -169]

TYPES OF RELATIONS

DIBY-45

Let A be the set of all functions $f: Z \rightarrow Z$ and R be a relation on A such that

$R = \{(f, g): f(0) = g(1) \text{ and } f(1) = g(0)\}$. Then R is:

[JEE Mains-2025]

- (A) Symmetric and transitive but not reflective
- (B) Symmetric but neither reflective nor transitive.
- (C) Reflexive but neither symmetric nor transitive.
- (D) Transitive but neither reflexive nor symmetric.

[Ans.: (B)]

DIBY-46

Let $A = \{-3, -2, -1, 0, 1, 2, 3\}$. Let R be a relation on A defined by xRy if and only if $0 \leq x^2 + 2y \leq 4$. Let l be the number of elements in R and m be the minimum number of elements required to be added in R to make it a reflexive relation.

Then $l + m$ is equal to:

[JEE Mains-2025]

- (A) 19
- (B) 20
- (C) 17
- (D) 18

[Ans.: (D)]

TYPES OF RELATIONS

DIBY-47

Let $A = \{-2, -1, 0, 1, 2, 3, 4\}$. Let R be a relation on A defined by xRy if and only if $2x + y \leq 2$. Let l be the number of elements in R . Let m and n be the minimum number of elements required to be added in R to make it reflexive and symmetric relations respectively. Then $l + m + n$ is equal to: [JEE Mains-2026]

- (A) 35 (B) 34 (C) 33 (D) 32

[Ans. : (C)]

DIBY-48

Let the relations R_1 and R_2 on the set $X = \{1, 2, 3, \dots, 20\}$ be given by

$R_1 = \{(x, y) : 2x - 3y = 2\}$ and $R_2 = \{(x, y) : -5x + 4y = 0\}$. If M and N be the minimum number of elements required to be added in R_1 and R_2 , respectively, in order to make the relations symmetric; then $M + N$ equals: [JEE Mains-2024]

[Ans. : 10]

DIBY-49

Let $A = \{1, 2, 3, \dots, 100\}$. Let R be a relation on A defined by $(x, y) \in R$ if and only if $2x = 3y$. Let R_1 be a symmetric relation on A such that $R \subset R_1$ and the number of elements in R_1 is n . Then, the minimum value of n is ____ . [JEE Mains-2024]

[Ans. : 66]

FUNCTIONS

DIBY-50

SOLVING INEQUALITY (WARM-UP)

Solve the following inequalities :

$$(i) \frac{5}{2x-1} \geq 0$$

$$(ii) x(3-x)(x-5) > 0$$

$$(iii) \frac{x(x-2)}{5(x^2-9)} \geq 0$$

$$(iv) \frac{x+1}{x-1} \geq \frac{x+5}{x-1}$$

$$(v) \frac{x^2-5x+6}{x^2-x-12} \leq 0$$

[Ans: (i) $x \in \left(\frac{1}{2}, \infty\right)$
(ii) $x \in (-\infty, 0) \cup (3, 5)$
(iii) $x \in (-\infty, -3) \cup [0, 2] \cup (3, \infty)$
(iv) $x \in (-\infty, 1)$
(v) $x \in (-3, 2] \cup [3, 4)$]

DOMAIN OF FUNCTION



DIBY-51

Domain of $f(x) = \frac{\sqrt{4-x^2}}{x^2-x}$

Ans. $x \in [-2, 2] - \{0, 1\}$

DIBY-52

The domain of definition of $f(x) = \frac{\log_2(x+3)}{x^2+3x+2}$ is

[JEE 2001, 1M]

(A) $R - \{-1, -2\}$

(B) $(-2, \infty)$

(C) $R - \{-1, -2, -3\}$

(D) $(-3, \infty) - \{-1, -2\}$

Ans. D

DIBY-53

The domain of definition of $y = \frac{1}{\log_{10}(1-x)} + \sqrt{x+2}$ is

[JEE 1983, 1M]

(A) $(-3, -2)$ excluding -2.5

(B) $[0, 1]$ excluding 0.5

(C) $[-2, 1)$ excluding 0

(D) None of these

Ans. C

DOMAIN OF FUNCTION

DIBY-54

Let $y = \sqrt{\frac{(x+1)(x-3)}{(x-2)}}$. Find all the real values of x for which y takes real values.

[JEE 1980, 2M]

Ans. $x \in [-1, 2) \cup [3, \infty)$

DIBY-55

The domain of the function $f(x) = \sqrt{(x-1)} + \sqrt{(2-x)} + \frac{1}{2x-3}$ is

- | | |
|--------------|---|
| (A) $[1, 2]$ | (B) $[1, 2] - \left\{\frac{3}{2}\right\}$ |
| (C) $(1, 2)$ | (D) $\left(1, \frac{3}{2}\right)$ |

Ans. B

DOMAIN OF FUNCTION

DIBY-56

Domain of $F(x) = \frac{1}{\sqrt{x-4}-\sqrt{6-x}}$

[Ans. : $[4, 6] - \{5\}$]

DIBY-57

The complete solution set of the inequality $\frac{3^x(2x-5)(x^2+x+2)}{(\cos x-2)(x^2+x)} \leq 0$ is

- (A) $(-\infty, -1)$ (B) $(\frac{5}{2}, \infty)$ (C) $(-1, \frac{5}{2}]$ (D) $(-1, 0) \cup [\frac{5}{2}, \infty)$
- [Ans. (D)]

DIBY-58

Domain of the function $f(x) = \frac{1}{\log_e(9-x^2)}$

[Ans. : $(-3, 3) - \{-2\sqrt{2}, 2\sqrt{2}\}$]

DOMAIN OF FUNCTION

DIBY-59

The domain of definition of the function $y = \frac{1}{\log_{10}(1-x)} + \sqrt{x+2}$ is

[Ans. $(-2, 1) - \{0\}$]

DIBY-60

The domain of the function, $f(x) = \frac{\sqrt{\sin x}}{\sqrt{(x-2)(8-x)}}$ is

(A) $[0, \pi] \cup [2\pi, 8)$

(B) $(2, \pi] \cup [2\pi, 8)$

(C) $(2, 8)$

(D) $(0, 8)$

[Ans. (B)]

DIBY-61

If the domain of the function $f(x) = \log_e \left(\frac{2x+3}{4x^2+x-3} \right) + \cos^{-1} \left(\frac{2x-1}{x+2} \right)$ is $(\alpha, \beta]$, then the value of $5\beta - 4\alpha$ is equal to

[JEE Mains-2024]

(A) 10

(B) 12

(C) 11

(D) 9

[Ans. (B)]

DOMAIN OF FUNCTION

DIBY-62

If the domain of the function $f(x) = \frac{\sqrt{x^2-25}}{(4-x^2)} + \log_{10}(x^2 + 2x - 15)$ is $(-\infty, \alpha) \cup [\beta, \infty)$, then $\alpha^2 + \beta^3$ is equal to:

[JEE Mains-2024]

- (A) 14 (B) 175 (C) 150 (D) 125

[Ans.: (C)]

DIBY-63

If the domain of the function $\sin^{-1}\left(\frac{3x-22}{2x-19}\right) + \log_e\left(\frac{3x^2-8x+5}{x^2-3x-10}\right)$ is $(\alpha, \beta]$, then $3\alpha + 10\beta$ is equal to:

[JEE Mains-2024]

- (A) 97 (B) 100 (C) 95 (D) 98

[Ans.: (A)]

DIBY-64

If domain of the function $\log_e\left(\frac{6x^2 + 5x + 1}{2x - 1}\right) + \cos^{-1}\left(\frac{2x^2 - 3x + 4}{3x - 5}\right)$ is $(\alpha, \beta) \cup (\gamma, \delta]$, then $18(\alpha^2 + \beta^2 + \gamma^2 + \delta^2)$ is equal to _____.

[JEE Mains-2023]

[Ans.: 20]

DOMAIN OF FUNCTION

DIBY-65

The domain of the function $f(x) = \sin^{-1} \left(\frac{x^2 - 3x + 2}{x^2 + 2x + 7} \right)$ is:

- (A) $[1, \infty)$ (B) $[-1, 2]$ (C) $[-1, \infty)$ (D) $(-\infty, 2]$

[Ans.: (C)]

DIBY-66

The domain of the function $f(x) = \sin^{-1} \left(\frac{3x^2 + x - 1}{(x - 1)^2} \right) + \cos^{-1} \left(\frac{x - 1}{x + 1} \right)$ is:

[JEE Mains-2021]

- (A) $\left[\frac{1}{4}, \frac{1}{2} \right] \cup \{0\}$ (B) $\left[0, \frac{1}{2} \right]$ (C) $\left[0, \frac{1}{4} \right]$ (D) $[-2, 0] \cup \left[\frac{1}{4}, \frac{1}{2} \right]$

[Ans.: (A)]

DIBY-67

If the domain of the function $f(x) = \sin^{-1} \left(\frac{5-x}{3+2x} \right) + \frac{1}{\log_e(10-x)}$ is $(-\infty, \alpha] \cup [\beta, \gamma) - \{\delta\}$,

then $6(\alpha + \beta + \gamma + \delta)$ is equal to:

[JEE Mains-2026]

[Ans.: 70]

DOMAIN OF FUNCTION

DIBY-68

If the domain of the function $f(x) = \sec^{-1}\left(\frac{2x}{5x+3}\right)$ is $[\alpha, \beta) \cup (\gamma, \delta]$, then $|3\alpha + 10(\beta + \gamma) + 21\delta|$ is equal to

[JEE Mains-2023]

[Ans. : 24]

DIBY-69

Let the domain of the function $f(x) = \cos^{-1}\left(\frac{4x+5}{3x-7}\right)$ be $[\alpha, \beta]$ and the domain of $g(x) = \log_2(2 - 6\log_{27}(2x + 5))$ be (γ, δ) . Then $|7(\alpha + \beta) + 4(\gamma + \delta)|$ is equal to ____.

[JEE Mains-2025]

[Ans. : 96]

DIBY-70

If the domain of the function:

$f(x) = \log_e(4x^2 + 11x + 6) + \sin^{-1}(4x + 3) + \cos^{-1}\left(\frac{10x+6}{3}\right)$ is $(\alpha, \beta]$, then $36|\alpha + \beta|$ is equal to:

[JEE Mains-2023]

[Ans. : 45]

*It is never about end result
It's always about Journey*



THANK
You

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